

SkillNav: A Multi-Agent Blackboard for Hallucination-Resistant Zero-Shot Object Navigation

Zhi Yang¹ Chengpu Yu^{1,2*} Yipin Zheng¹

¹School of Automation, Beijing Institute of Technology, Beijing 100081, China

²Chongqing Innovation Center, Beijing Institute of Technology, Chongqing 401147, China
{3220241334, yuchengpu, zhengyipin}@bit.edu.cn

Abstract

Embodied agents that route every high-level decision through a single language model typically ingest the entire trajectory at every step, a regime in which prompt context grows monotonically and both *lost-in-the-middle* degradation and trajectory-level hallucination peak. We show that *multi-agent decoupling* on a shared *Voronoi-Frontier blackboard* is a structural answer. SkillNav exposes a sparse topology over the partially-built map whose nodes carry visual evidence and two multiplicative gates ($\mu_{\text{safe}}, \mu_{\text{mem}}$) written by specialised Safe and Memory agents; a strategic Exploration agent reads only a compact, system-prompt-style summary of the blackboard — whose size is bounded by the topology rather than by trajectory length — and writes back only a phase decision plus an SR/IG weight pair. The three agents never see one another’s prompts; false-positive evidence is suppressed *architecturally* by μ_{mem} rather than by asking the language model to revise its own past output. On HM3D val, SkillNav reaches **71.9 SR / 34.0 SPL** with only **22.5** LLM calls per episode ($\sim 11.9\text{K}$ total tokens), surpassing the strongest zero-shot baseline in our comparison (RATE-Nav) by $+4.1$ SR / $+2.7$ SPL.

1 Introduction

Embodied object goal navigation (ObjectNav) (Chaplot et al., 2020; Savva et al., 2019) asks an agent to locate a target category specified at test time in an unseen environment, using only first-person RGB-D and odometry. In the *zero-shot* setting (Majumdar et al., 2022) no per-category training is permitted, so the system must lean entirely on the priors carried by a frozen vision-language foundation model and a strategic language model that decides where to go next. The dominant recipe today is to consult that LM at every step, feeding it the trajectory, the

visible candidates, and the topology summary all together (Wu et al., 2024a; Yu et al., 2023; Zhang et al., 2024).

This per-step recipe is structurally hallucination-prone. Prompt context grows monotonically with episode length; *lost-in-the-middle* degradation (Liu et al., 2024a) peaks exactly when the agent most needs to recall earlier evidence; and a single confidently-wrong false-positive (e.g., a vase mistaken for a potted plant) compounds across hundreds of subsequent decisions, an effect documented for LLM-driven embodied agents in particular (Chakraborty et al., 2025; Lin et al., 2025). The standard remedy — *grounding* the LM in an external, structured state — is well-explored for retrieval and knowledge-base settings but largely untouched in embodied navigation, where the “external state” is a partial, dynamically-built map and the “groundable evidence” is the agent’s own VLM observations.

We argue that grounding an LM for embodied decision-making is best done through a *shared spatial blackboard* rather than a richer prompt. SkillNav (fig. 1) exposes a Voronoi-Frontier graph in which every node v carries a small VLM-derived memory slot (room hypothesis, observed objects, freshness) and two multiplicative gates: $\mu_{\text{safe}}(v)$ written by a Safe agent that owns geometric viability, and $\mu_{\text{mem}}(v)$ written by a Memory agent that owns multi-frame verification. A strategic LM (DeepSeek-Chat) operates on a compact, system-prompt-style view of the blackboard — the per-call input is dominated by a static system prefix that the VLM-driven Memory agent keeps up to date in the background, rather than by an ever-growing trajectory transcript — and emits only a phase label and a $(w_{\text{SR}}, w_{\text{IG}})$ fusion-weight pair, never the trajectory plan itself. False-positive evidence is suppressed *architecturally* (μ_{mem} multiplicatively decays toward a floor), not by asking the LM to revise its own past output.

*Corresponding author: yuchengpu@bit.edu.cn.

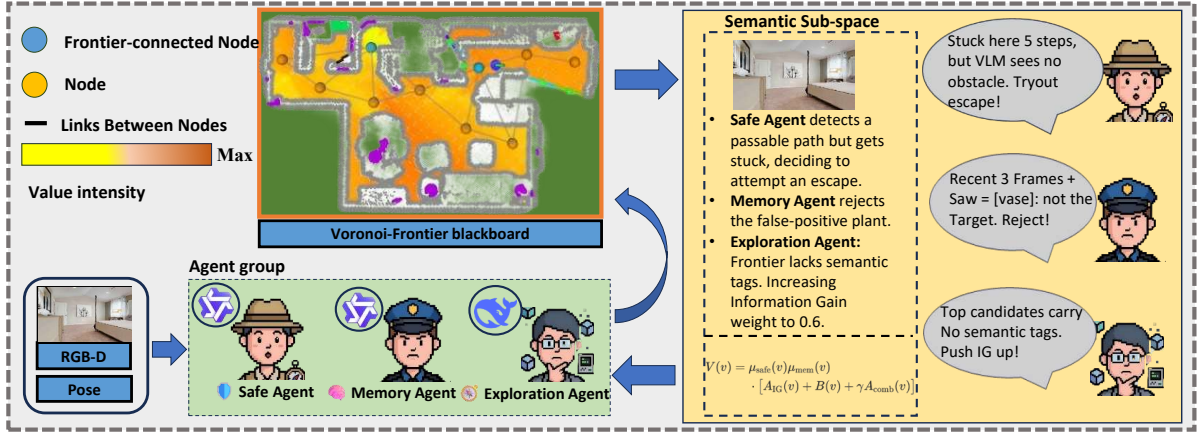


Figure 1: **SkillNav grounds a strategic LM in a Voronoi-Frontier blackboard.** **Left:** RGB-D + pose drives the blackboard — decision nodes (yellow / blue when anchoring frontiers) connected by reachability edges, with per-cell value intensity shown by the colour bar. **Right:** three agents reason in their own narrow semantic sub-spaces and write disjoint channels back — Safe (μ_{safe} , “Stuck here 5 steps, but VLM sees no obstacle. Tryout escape!”), Memory (μ_{mem} , “Recent 3 frames + saw=[vase]: not the target. Reject!”), and Exploration (w_{SR}, w_{IG} , “Top candidates carry no semantic tags. Push IG up!”). The blackboard’s algebra $V(v) = \mu_{safe}\mu_{mem}[A_{IG} + B + \gamma A_{comb}]$ combines the updates from the three agents.

Contributions. (1) *A multi-agent blackboard that decouples context:* three role-specialised agents read the same per-node state (μ_{safe} , μ_{mem} , memory slot) but never read one another’s prompts, keeping the strategic LM’s prompt size independent of episode length (§3.1). (2) *Architectural hallucination suppression:* a multiplicative gate μ_{mem} that compounds VLM verdicts over repeated visits, suppressing persistent false-positive frontiers without LM self-correction (§3.2). (3) *Bounded-cost grounded inference:* on HM3D val, SkillNav reaches **71.9 SR / 34.0 SPL** with only **22.5 LM** calls per episode and $\sim 11.9K$ total tokens, surpassing the strongest zero-shot baseline in our comparison (RATE-Nav) by $+4.1$ SR / $+2.7$ SPL (§4).

2 Related Work

Zero-shot Object Navigation. Frontier-based ZSON systems score each frontier cell with a single VLM signal and pick the best at every step (VLFM, L3MVN, ESC (Yokoyama et al., 2024; Yu et al., 2023; Zhou et al., 2023)). TriHelper (Zhang et al., 2024) adds three helper modules sharing one planner; ImagineNav (Zhao et al., 2025) prompts a VLM to imagine future views; RATE-Nav (Li et al., 2025) adds region-aware termination. Topology methods reason at the node level instead: VoroNav (Wu et al., 2024a) runs an LLM at every Voronoi waypoint, SG-Nav (Yin et al., 2024) prompts over a 3D scene graph, CogNav (Cao et al., 2025) casts the episode as a 5-state LLM-driven

process. All of them route every decision through one LM with a shared, monotonically-growing context.

Grounding and hallucination. Grounding has been studied extensively for text and knowledge bases (Lewis et al., 2020), but these works ground the LM in *retrieved text*, not in a dynamically-built spatial state. Long-context LMs lose middle-context information (Liu et al., 2024a); LLM-based embodied agents compound this into trajectory-level hallucination (Chakraborty et al., 2025; Lin et al., 2025). Common mitigations (self-revision, RAG over prior outputs) still ask the LM to correct its own past assertions; SkillNav suppresses false-positive evidence *below* the LM, in the blackboard’s algebra.

Multi-agent LLM architectures. Recent agentic systems (Wu et al., 2024b; Park et al., 2023) coordinate via free-form dialogue, recreating the long-context problem at the inter-agent level. SkillNav inherits the *blackboard* pattern from classical problem solving (Erman et al., 1980): agents write to disjoint channels, never to each other’s prompts. Combined with the options framework (Sutton et al., 1999) this yields strict context isolation.

3 Method

3.1 The Voronoi-Frontier Blackboard

Given a partially-built occupancy map at time t , we extract a sparse Voronoi graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ over ex-

explored free space; nodes are degree- ≥ 3 junctions and edges are reachability paths between them. The current frontier set $\mathcal{F}$ is anchored to nodes by an A^* -aware reachability map $\pi : \mathcal{F} \rightarrow \mathcal{V} \cup \{\perp\}$; orphans ($\pi(f) = \perp$) feed a fallback branch (App. D).

Every node carries two BLIP-2 ITM scores $V_{\text{SR}}^{\text{cell}}(x_v), V_{\text{IG}}^{\text{cell}}(x_v)$ under two complementary prompts (App. C), a VLM memory slot (room hypothesis, observed objects, freshness) refreshed intermittently by the Memory agent, and two scalar gates $\mu_{\text{safe}}(v), \mu_{\text{mem}}(v) \in [\mu_{\min}, 1]$ written by the Safe and Memory agents. The strategic LM writes only the fusion weights ($w_{\text{SR}}, w_{\text{IG}}$).

At every decision tick, node selection maximises a unified value $v^* = \arg \max_v V(v)$, where

$$V(v) = \mu_{\text{safe}}(v) \mu_{\text{mem}}(v) \cdot [A_{\text{IG}}(v) + B(v) + \gamma A_{\text{comb}}(v)], \quad (1)$$

clipped to $[V_{\min}, V_{\max}] = [0.01, 1.0]$. The three additive terms are

$$A_{\text{IG}}(v) = \sigma^{-1} \sum_{f \in \pi^{-1}(v)} \text{IG}_{\text{geom}}(f), \quad (2)$$

$$B(v) = \beta \log(1 + \deg(v)), \quad (3)$$

$$A_{\text{comb}}(v) = w_{\text{SR}} V_{\text{SR}}^{\text{cell}} + w_{\text{IG}} V_{\text{IG}}^{\text{cell}}, \quad (4)$$

where A_{IG} aggregates the anchored geometric information gain, B is a connectivity bonus (Wu et al., 2024a), and A_{comb} is the LM-weighted dual-ValueMap score. Because μ_{safe} and μ_{mem} are independent channels, writes commute and eq. (1) is order-invariant — this makes context isolation *architectural* rather than conventional.

3.2 Three Agents on Disjoint Channels

Table 1 summarises which channel each agent owns and how it reasons. All three agents read the same blackboard state; they never see one another’s prompts.

Memory. On each VLM verdict at v , the memory gate is updated as

$$\mu_{\text{mem}} \leftarrow \text{clip}(\gamma_e \mu_{\text{mem}}, \mu_{\min}, \mu_{\max}), \quad (5)$$

with $\gamma_e \in \{0.5, 0.7, 1.1\}$ for a multi-frame FP, a single-shot FP, and a TP respectively, and floor $\mu_{\min} = 0.1$. Three repeated REJECTs take μ_{mem} to 0.125, dropping $V(v)$ by an order of magnitude regardless of SR/IG — this compounding form is the architectural hallucination suppressor.

Safe. Tryout escape writes

$$\mu_{\text{safe}} \leftarrow \max(\mu_{\min}, \gamma_{\text{stuck}} \mu_{\text{safe}}) \quad (6)$$

on each consecutive escape failure ($\gamma_{\text{stuck}} = 0.3$, floor 0.05); three failures would yield ≈ 0.027 before clipping and therefore reach the 0.05 floor, encoding a geometric blocker as a language-agnostic gate. The Safe agent never calls a language model.

Exploration. A strategic LM (DeepSeek-Chat) is consulted every 3 s with a compact state vector (coverage, candidate count, FP count, stuck count, previous phase) and returns JSON with the next phase and a normalised ($w_{\text{SR}}, w_{\text{IG}}$). A rule-based fallback mirrors the LM’s tiers to allow navigation to continue when the LM API is unavailable (App. C).

4 Experiments

4.1 Setup

We evaluate on the HM3D ObjectNav val split (Ramakrishnan et al., 2021) in the standard $25 \times 25 \text{ m}^2$ configuration with a 500-step horizon and 1 m success radius. SR and SPL are computed as in Savva et al. (2019). SkillNav is implemented on a Habitat-ROS bridge; the strategic LM is DeepSeek-Chat consulted at a fixed 3 s cadence, the Memory verifier is Qwen-VL-Plus, and both SR and IG channels score BLIP-2 ITM (Li et al., 2023) under the templates in App. C. All hyperparameters (App. B) are tuned once on HM3D train and fixed across evaluations.

4.2 Main Results

Table 2 reports HM3D val SR and SPL. SkillNav reaches **71.9 SR / 34.0 SPL**, +4.1 SR / +2.7 SPL over the strongest zero-shot baseline in our comparison (RATE-Nav, ACL’25 (Li et al., 2025)). It also surpasses the privileged *ImagineNav-Oracle* variant (Zhao et al., 2025) by +9.9 SR / +2.9 SPL.

4.3 Ablation Study

Table 3 evaluates the contributions of the Safe and Memory agents and adaptive SR/IG control on the same **200** HM3D-v2 val episodes. The full model achieves 72.5 SR / 36.3 SPL on this subset.

Safe agent. Removing the Safe agent yields the largest decrease in both metrics among the evaluated ablations: SR falls by 10.5 points and SPL by 3.2 points relative to the full model. Stuck-related terminal outcomes (including their false-negative

Agent	Reads	Writes	Vocabulary	Backbone
Safe	pose, occupancy, recent trajectory	$\mu_{\text{safe}}(v)$	{wedged, stuck, tryout, ok}	pure geometry (no LM)
Memory	multi-frame RGB + per-cand. FP count	$\mu_{\text{mem}}(v)$	{CONFIRM, UNCERTAIN, REJECT}	Qwen-VL-Plus, event-trig.
Exploration	compact blackboard summary	$(w_{\text{SR}}, w_{\text{IG}})$, phase	{BROAD, DIRECTED, APPROACH, VERIFY}	DeepSeek, 3 s cadence

Table 1: The three agents of SkillNav. Each reads the shared blackboard but writes a single, named channel; vocabularies are disjoint so writes commute. Full prompts in App. C.

Method	SR $\uparrow$	SPL $\uparrow$
ESC (Zhou et al., 2023)	39.2	22.3
L3MVN (Yu et al., 2023)	48.7	23.0
VLFM (Yokoyama et al., 2024)	52.4	30.3
VoroNav (Wu et al., 2024a)	42.0	26.0
TriHelper (Zhang et al., 2024)	56.5	25.3
InstructNav (Long et al., 2024)	58.0	20.9
OpenFMNav (Kuang et al., 2024)	52.5	24.1
SG-Nav (Yin et al., 2024)	54.2	24.1
ImagineNav (Zhao et al., 2025)	53.0	23.8
ImagineNav-Oracle [†] (Zhao et al., 2025)	62.0	31.1
RATE-Nav (Li et al., 2025)	67.8	31.3
SkillNav (ours)	71.9	34.0

Table 2: HM3D val results. [†] Privileged oracle variant with real panoramic captures at candidate poses.

variants) increase from 1 to 24 episodes. These results support the contribution of the Safe agent to handling local motion failures on this subset. The agent encodes motion and occupancy evidence in μ_{safe} , allowing the planner to downweight nodes associated with repeated escape failures.

Memory agent. Removing the Memory agent lowers SR by 3.5 points and SPL by 3.1 points relative to the full model. By retaining verification evidence across views, the Memory agent carries corrective information into subsequent decisions and downweights rejected candidates through μ_{mem} . This mechanism helps limit the persistent influence of false-positive hallucinations, where non-target objects are mistaken for the target. The decreases in SR and SPL support the contribution of memory to navigation performance on this subset.

Exploration agent: adaptive SR/IG control. The Exploration agent adaptively adjusts SR/IG weights to prioritise likely target locations. This may encourage premature commitment to a small set of candidates, limiting the alternatives presented to the Memory agent for verification and prolonging pursuit of misidentified objects. This is consistent with the increase in false-positive terminal

Mem	Safe	Expl	SR $\uparrow$	SPL $\uparrow$
×	✓	✓	69.0	33.2
✓	×	✓	62.0	33.1
✓	✓	×	73.5	35.5
✓	✓	✓	72.5	36.3

Table 3: Ablations on the same 200-episode HM3D-v2 val subset. Mem/Safe ×: remove the Memory/Safe agent, respectively. Expl ×: fix SR/IG weights, retaining other Exploration functions. ✓: retain the component. All checks: full model.

outcomes from 16 with fixed weights to 22 with adaptive weights, accompanied by a decrease in SR from 73.5 to 72.5. Nevertheless, SPL increases from 35.5 to 36.3 despite the lower SR, indicating improved success-weighted path efficiency on this subset.

4.4 Strategic LM Inference Cost

Table 4 reports the measured per-episode cost of the strategic LM in SkillNav on the HM3D val sweep. Each call’s prompt is dominated by a static system prefix plus a compact blackboard snapshot, so its size is set by the topology rather than by trajectory length. Per-step LLM designs such as VoroNav (Wu et al., 2024a) or L3MVN (Yu et al., 2023) consult the LM once per low-level decision; on a 500-step episode this is upper-bounded by ~ 500 calls by construction. SkillNav’s option-boundary cadence cuts this to 22.5 calls per episode while *improving* SR (71.9 vs. VoroNav’s 42.0 in table 2).

Figure 2 decomposes the headline numbers by HM3D category: large-furniture targets clear 75% SR while small or partially-occluded targets cluster near 55%, a pattern consistent with prior ObjectNav benchmarks (Yokoyama et al., 2024; Zhang et al., 2024); the dominant remaining failure mode is detection rather than navigation. A full failure-mode taxonomy is given in App. A.

Metric (HM3D val)	Value
LM calls / episode (mean)	22.5
Prompt tokens / call (mean)	527
Total tokens / episode (mean)	11,885
LM call latency (mean / p_{95})	1.95 s / 2.50 s
VLM verifier calls / episode	4.54

Table 4: Measured per-episode cost of the strategic LM (DeepSeek-Chat) in SkillNav on the HM3D val sweep. Token counts are for the strategic LM only; the upstream VLM cost (Memory verifier ~ 4.5 calls/ep, BLIP-2 ITM scoring of the dual ValueMap) is not amortised into this prompt and is accounted for separately.

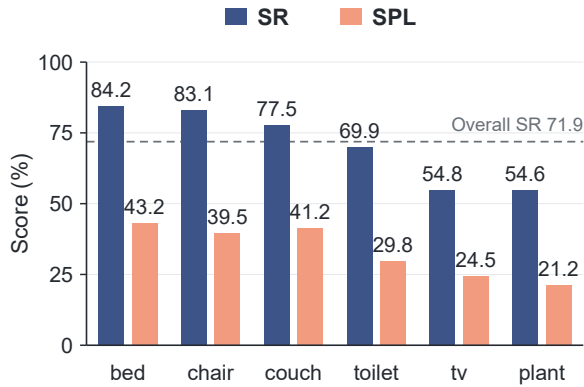


Figure 2: Per-category SR and SPL on HM3D val. The dashed line marks the overall SR (71.9).

4.5 Context Isolation and Hallucination Resistance

A monolithic agent serialises observations, decisions, and recovery attempts into a growing trajectory prompt. SkillNav instead stores typed fields: Memory writes its verification state and μ_{mem} , Safe writes μ_{safe} , and Exploration reads only the top- K nodes and candidates. Earlier visual and motion evidence therefore remains actionable without repeated replay.

At 527 tokens per call and 22.5 calls per episode (table 4), the strategic context need not grow with the trajectory. An early rejection or stuck event remains an explicit value, while channel ownership prevents one agent from erasing another’s evidence. This bounded interface reduces exposure to lost-in-the-middle behaviour.

Channel ownership also makes interventions auditable: Memory downweights rejected targets, Safe records repeated motion, and Exploration cannot overwrite either verdict. This shortens context, although top- K may omit a useful distant frontier. Each error can also be traced to a named channel

without reconstructing the complete action history.

The full run still has 83 false-positive lock-ins (8.3%; App. A), and the Memory ablation does not prove a lower terminal false-positive rate. The results support preservation of corrective signals; direct frame-level hallucination tests and a matched monolithic baseline are left to future work.

5 Conclusion

We ground a strategic LM in a Voronoi–Frontier blackboard carrying VLM evidence and gates written by Safe and Memory agents. The LM reads a topology-bounded view and writes only fusion weights and a phase; μ_{mem} suppresses false-positive evidence. SkillNav reaches 71.9 SR / 34.0 SPL on HM3D val with 22.5 LM calls per episode.

On the evaluated ablation subset, removing the Safe agent produces the largest decrease in SR and SPL, together with more stuck-related terminal outcomes. Removing the Memory agent also lowers both metrics. Adaptive SR/IG weighting yields higher SPL but lower SR than fixed weights. These results support the contributions of the individual components to navigation performance on this subset.

Limitations

- (i) Experiments use the HM3D validation split in Habitat and one VLM/LM stack; cross-simulator and cross-model generalisation remain unverified.
- (ii) Intermittent memory updates can lag the current pose.
- (iii) The fixed-size LM summary saves tokens but omits graph-global structure that a graph-attention head could exploit.

Ethics Statement

SkillNav is evaluated only in HM3D simulation, with no human subjects or private data. DeepSeek and DashScope APIs are used under their standard terms. We see no direct dual-use risk beyond embodied navigation research.

Acknowledgments

This work was supported by the National Natural Science Foundation of China under Grants 62473046, T2541018, 72642006 and 62688201.

References

- Yihan Cao, Jiazhao Zhang, Zhinan Yu, Shuzhen Liu, Zheng Qin, Qin Zou, Bo Du, and Kai Xu. 2025. CogNav: Cognitive process modeling for object goal navigation with LLMs. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.
- Trishna Chakraborty, Udit Ghosh, Xiaopan Zhang, Fahim Faisal Niloy, Yue Dong, Jiachen Li, Amit K. Roy-Chowdhury, and Chengyu Song. 2025. HEAL: An empirical study on hallucinations in embodied agents driven by large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. ArXiv:2506.15065.
- Devendra Singh Chaplot, Dhiraj P. Gandhi, Abhinav Gupta, and Russ R. Salakhutdinov. 2020. Object goal navigation using goal-oriented semantic exploration. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 4247–4258.
- Lee D. Erman, Frederick Hayes-Roth, Victor R. Lesser, and D. Raj Reddy. 1980. The Hearsay-II speech-understanding system: Integrating knowledge to resolve uncertainty. *ACM Computing Surveys*, 12(2):213–253.
- Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. 2023. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4015–4026. IEEE.
- Yuxuan Kuang, Hai Lin, and Meng Jiang. 2024. OpenFMNav: Towards open-set zero-shot object navigation via vision-language foundation models. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 338–351, Mexico City, Mexico. Association for Computational Linguistics.
- Boris Lau, Christoph Sprunk, and Wolfram Burgard. 2010. Improved updating of Euclidean distance maps and Voronoi diagrams. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 281–286. IEEE.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Junjie Li, Nan Zhang, Xiaoyang Qu, Kai Lu, Guokuan Li, Jiguang Wan, and Jianzong Wang. 2025. RATE-Nav: Region-aware termination enhancement for zero-shot object navigation with vision-language models. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 6564–6574, Vienna, Austria. Association for Computational Linguistics.
- Junnan Li, Dongxu Li, Silvio Savarese, and Steven C. H. Hoi. 2023. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, pages 19730–19742. PMLR.
- Xixun Lin, Yucheng Ning, Jingwen Zhang, Yan Dong, Yilong Liu, Yongxuan Wu, Xiaohua Qi, Nan Sun, Yanmin Shang, Kun Wang, Pengfei Cao, Qingyue Wang, Lixin Zou, Xu Chen, Chuan Zhou, Jia Wu, Peng Zhang, Qingsong Wen, Shirui Pan, and 5 others. 2025. LLM-based agents suffer from hallucinations: A survey of taxonomy, methods, and directions. *arXiv preprint arXiv:2509.18970*.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024a. [Lost in the middle: How language models use long contexts](#). *Transactions of the Association for Computational Linguistics*, 12:157–173.
- Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. 2024b. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 38–55. Springer.
- Yuxing Long, Wenzhe Cai, Hongchen Liu, Fangwei Zhan, and He Wang. 2024. InstructNav: Zero-shot system for generic instruction navigation in unexplored environments. In *Proceedings of the 8th Conference on Robot Learning (CoRL)*, pages 1–16.
- Arjun Majumdar, Gunjan Aggarwal, Bhavika Devnani, Judy Hoffman, and Dhruv Batra. 2022. ZSON: Zero-shot object-goal navigation using multimodal goal embeddings. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 32340–32352.
- Joon Sung Park, Joseph C. O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative agents: Interactive simulators of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*.
- Santhosh K. Ramakrishnan, Aaron Gokaslan, Erik Wijmans, Oleksandr Maksymets, Alex Clegg, John Turner, Eric Undersander, Wojciech Galuba, Andrew Westbury, Angel X. Chang, Manolis Savva, Yili Zhao, and Dhruv Batra. 2021. Habitat-Matterport 3d dataset (HM3D): 1000 large-scale 3d environments for embodied AI. In *Proceedings of the Neural Information Processing Systems (NeurIPS) Track on Datasets and Benchmarks*.
- Manolis Savva, Abhishek Kadian, Oleksandr Maksymets, Yili Zhao, Erik Wijmans, Bhavana Jain, Julian Straub, Jia Liu, Vladlen Koltun, Jitendra Malik, Devi Parikh, and Dhruv Batra. 2019. Habitat:

A platform for embodied AI research. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9339–9347. IEEE.

Richard S. Sutton, Doina Precup, and Satinder Singh. 1999. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1–2):181–211.

Pengying Wu, Yao Mu, Bingxian Wu, Yi Hou, Ji Ma, Shanghang Zhang, and Chang Liu. 2024a. VoroNav: Voronoi-based zero-shot object navigation with large language model. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*. PMLR.

Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W. White, Doug Burger, and Chi Wang. 2024b. AutoGen: Enabling next-gen LLM applications via multi-agent conversation framework. In *Proceedings of the 1st Conference on Language Modeling (COLM)*.

Hang Yin, Xiuwei Xu, Zhenyu Wu, Jie Zhou, and Jiwen Lu. 2024. SG-Nav: Online 3d scene graph prompting for llm-based zero-shot object navigation. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Naoki Yokoyama, Sehoon Ha, Dhruv Batra, Jiuguang Wang, and Bernadette Bucher. 2024. VLFM: Vision-language frontier maps for zero-shot semantic navigation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1–8. IEEE.

Bangguo Yu, Hamidreza Kasaei, and Ming Cao. 2023. L3MVN: Leveraging large language models for visual target navigation. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 3554–3560. IEEE.

Lingfeng Zhang, Qiang Zhang, Hao Wang, Erjia Xiao, Zixuan Jiang, Honglei Chen, and Renjing Xu. 2024. TriHelper: Zero-shot object navigation with dynamic assistance. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE.

Xinxin Zhao, Wenzhe Cai, Likun Tang, and Teng Wang. 2025. ImagineNav: Prompting vision-language models as embodied navigator through scene imagination. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Kaiwen Zhou, Kaizhi Zheng, Connor Pryor, Yilin Shen, Hongxia Jin, Lise Getoor, and Xin Eric Wang. 2023. ESC: Exploration with soft commonsense constraints for zero-shot object navigation. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, pages 42829–42842. PMLR.

A Full Failure-Mode Taxonomy

Automatic outcome breakdown over all **281** HM3D val failures (of 71.9% success):

- **Timed-out on correct path (61.2%** of failures, 17.2% of all episodes): the agent ran out of the 500-step budget along a feasible path to the goal; combines the `stepout_feasible` (10.0%) and `stepout_fn` (7.2%) outcome tags.
- **VLM false-positive locked (29.5%** of failures, 8.3% of episodes): the agent committed to a candidate that the Memory agent’s verification eventually rejected, but not in time to recover within the step budget.
- **Genuine unreachable / no-frontier (6.0%** of failures, 1.7% of episodes): the target is behind a closed door, on another floor, or the explored region was exhausted with no remaining frontier; no purely navigational policy could succeed (`infeasible + no_frontier(_fn)`).
- **Un-escaped dead-zone (3.2%** of failures, 0.9% of episodes): the Safe agent’s μ_{safe} decay isolated a region whose re-entry was later needed (`stucking + stucking_fn`).
- **Map error / SLAM drift (<1%)**: rare in Habitat with provided GT pose; included for taxonomy completeness.
- **Other / unclassified (0%)**.

B Hyperparameters

Table 5 lists every hyperparameter introduced in §3. All values are tuned once on HM3D train and held fixed across all benchmark splits.

C Prompt Templates

Semantic-relevance prompt. The SR channel uses a goal-conditioned BLIP-2 template:

“a {goal} in the room”

where {goal} is the episode’s object category (e.g. `toilet`, `potted plant`). The cosine similarity between this text and the current view is rendered into $V_{\text{SR}}^{\text{cell}}(x)$ across the agent’s projected field of view.

Symbol	Meaning	Value
<i>Frontier-anchored topology (§3.1)</i>		
λ	geom-IG weight in A_{IG}	0.5
β	degree-bonus weight	0.5
$r_{\max}$	anchor radius (A^*)	5.0 m
τ	orphan threshold (%-ile)	30
$d_{\deg}$	min node degree	3
<i>Memory agent (§3.2)</i>		
K_{ver}	verification frame window	5
γ_{fp}	per-FP penalty	0.1
λ_1	OppCost foregone weight	0.5
λ_2	OppCost FP-risk weight	1.0
<i>Safe agent (§3.2)</i>		
T_{stuck}	stuck window (ticks)	30
γ_{stuck}	per-failure μ decay	0.3
$\mu_{\min}$	multiplicative floor	0.05
<i>Exploration agent (§3.2)</i>		
Δt_{LM}	strategic LM cadence	3.0 s
$\rho_{\text{lo}}/\rho_{\text{hi}}$	coverage tiers	0.3/0.7
K	top-K nodes/candidates	8
<i>Habitat / ObjectNav (§4.1)</i>		
horizon	step budget	500
r_{success}	success radius	1.0 m
sensor	RGB-D, HFoV	79°

Table 5: Hyperparameters used in SkillNav. Tuned once on HM3D train; held fixed across all benchmarks and ablations.

Information-gain prompt. The IG channel uses a single, object-agnostic explorability template:

“An image of a doorway, corridor, or open passage leading to an unexplored region of the building.”

The template captures the classical explorability prior that doorways, corridors, and partial openings tend to reveal large unmapped areas. We experimented with multi-template averaging (three variants emphasising *doorway*, *corridor*, *opening* respectively); the gain over single-template was within ± 0.3 SR on HM3D val, and the single-template version is what we report in the main text.

Option-boundary LLM consultation. At each option-boundary tick or stagnation event, the Exploration agent sends a structured prompt to the language-model backend (DeepSeek-Chat) following the template in table 6. Each component is bounded in size: the topology summary lists at most $K = 8$ nodes, each formatted as $\langle \text{id}, \text{visited}, \text{nearby_objects}, V(v) \rangle$ in ~ 30 tokens; the candidate summary is similar; the total prompt stays under ~ 1.3 K tokens regardless of episode length.

System. You are the Exploration agent for a zero-shot object-goal navigation robot. Output the next navigation phase, and optionally an override of the SR / IG fusion weights. Respond in JSON with keys: `phase, w_sr, w_ig`.

User. Target object: `{goal}`.

Current phase: `{current_option}`.

Phase history (last 5): `[{o_{t-4}}, ..., {o_t}]`.

Top- K topology nodes (sorted by $V(v)$), each formatted as

`{id, visited, nearby=[obj_1, ...], V}`.

Top- K target candidates (sorted by reliability), each as `{obj, rel, fp, dist}`.

Choose the next phase from `{broad-explore, directed-search, target-approach, verification}`.

Table 6: Template for the strategic-LM consultation (§3.2). Slot values are populated from the current black-board state by the Exploration agent.

D Implementation Details

Source code. Source code is available at <https://github.com/Pandakingxabc/SkillNav>.

Habitat / ROS stack. SkillNav is implemented as a Python evaluator (`habitat_evaluation.py`) communicating with a C++ ROS planner (`exploration_node`) via standard ROS topics. The evaluator publishes RGB-D + odometry at ~ 3 Hz and the dual ITM scores at ~ 5 Hz; the planner publishes the current goal node, trajectory, and option state at ~ 1 Hz. Each episode is preceded by a hard reset of the planner process to ensure no cross-episode state leakage between scenes.

VLM serving. We host BLIP-2 ITM, Grounding-DINO (Liu et al., 2024b), and SAM (Kirillov et al., 2023) as independent GPU-backed Flask services listening on ports 12181–12184. The dual ITM is a single BLIP-2 endpoint queried twice per frame (once with the SR prompt, once with the IG prompt); we cache the visual encoder forward pass on the server side to amortise the cost.

Voronoi topology. The generalised Voronoi diagram is computed from the obstacle distance field at each map update (Lau et al., 2010). Decision nodes are extracted as Voronoi vertices with degree ≥ 3 (junctions), filtered by a minimum inter-node distance of 1.5 m and a minimum obstacle clearance of 0.3 m. The graph typically contains 30–100 nodes mid-episode in HM3D scenes.

Frontier extraction and anchoring. Frontiers are extracted as 8-connected boundaries between free and unknown cells, clustered into groups of ≥ 5 cells. The anchor map π (§3.1) is computed by running a multi-source A* from all Voronoi nodes simultaneously; cells inherit the closest node’s id when their reachable distance is $\leq r_{\max} = 5.0$ m, else they enter the orphan queue.

Token instrumentation. LLM call statistics are collected by a lightweight wrapper around each backend client; for DeepSeek we use the exact `prompt_tokens` and `completion_tokens` fields from `response.usage`, and for Ollama the exact `prompt_eval_count / eval_count` fields. VLM call latency is measured at the client side as wall-clock around the Flask request. Where exact token counts are unavailable we fall back to a character-based approximation (~ 4 chars / token for ASCII, ~ 2 for CJK), and this approximation is used for $< 1\%$ of the calls in our final runs.

Compute. All evaluation was performed on a single workstation with one NVIDIA GeForce RTX 4070 Ti GPU for the local VLM servers (BLIP-2 ITM, Grounding-DINO, SAM), one CPU thread per Habitat simulator instance, and the DeepSeek-Chat and DashScope (Qwen-VL-Plus) API endpoints for the strategic LM and Memory verifier respectively. Wall-clock per HM3D episode averaged 85.6 s (median 60.6 s) for SkillNav; the full 1000-episode HM3D val sweep completed in 23.8 h.